All three-ρ terms proportional to $\log(\vee/\Lambda)$, with c.c.: final form

One equation. The lines are numbered as in your list; $\mathcal{D}$ has the lines D1–D8.

- **Black:** exactly as you wrote it.
- **Red: removed**, kept only for reference. These terms are *not* part of the sum.

  F4 Group I, Row II: the α part of the $U(w)\mathbb{A}^{(3)}$ sub-terms. It multiplies $-\log(\vee/k^+)$, not $\log(\vee/\Lambda)$.

  F3 Group I, Row III, part 2: the γ, γ' terms with the edge kernel ϕ.

- **Blue: changed or added**.

  T1 Group III, Row I, line 15: $\rho^h(x') \to \rho^c(x')$.

  T2 Group III, Row II, line 13: $U^{hl}(y) \to U^{hl}(x)$.

  T3 Group I, Row II, part 1, α lines: the kernel $\mathcal{K}(x'-w)\cdot\mathcal{K}(y-z) \to \mathcal{K}(x-z)\cdot\mathcal{K}(y-z)$.

  F3 $\mathcal{D}$, lines D1–D8: added. D5–D8 are its c.c., with the charges in reverse order.

  c.c. The c.c. of an operator product reverses the order of the charges.

  order The anticommutator of the notes, $\rho\{\rho,\rho\}$ or $\{\rho,\rho\}\rho$, with half of your coefficient, in place of one ordering with the full coefficient. In the c.c. it is also reversed.

The final result is the black and blue lines. Their symmetrized three-ρ parts cancel exactly. Reordering them gives the two-ρ terms that, together with the genuine two-ρ terms and those of `4rho.tex`, are JIMWLK⊗LO.

The order of the charges only matters for the two-ρ terms; for the three-ρ cancellation any order gives the same result. In Group I, Row II the two orderings of a pair are exactly $\rho\{\rho,\rho\}$ through the antisymmetry of f^{abc}, so the amplitude is kept as written and only the c.c. is reversed.

$$\frac{d^3N^{3\rho}_{\rm NLO}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{ig^4}{16\pi^5}\,\frac{1}{k^+}\,\log\left[\frac{\vee}{\Lambda}\right]\int_{x,y,z,x',w,w'} e^{-ik(w'-w)}\,\Bigg[$$

From four ρ's, d^3N_a (its overall minus sign is absorbed in the signs of its terms)

(1) $-\,\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x-z)\cdot\mathcal{K}(y-z)\,f^{abc}\rho^d(x')\rho^a(y)\rho^c(x)\,U^{de}(w')U^{eb}(w)$ (α)

(2) $+\,\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x-z)\cdot\mathcal{K}(y-z)\,f^{abc}\rho^e(x')\rho^a(y)\rho^c(x)\,U^{db}(w)U^{ed}(x')$ (α)

(3) $-\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\rho^c(x')\rho^a(y)\rho^e(x)\,U^{db}(w')U^{ed}(x)$ (α')

(4) $+\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\rho^c(x')\rho^a(y)\rho^e(x)U^{bd}(w')U^{ed}(w)$ (α')

From four ρ's, d^3N_b

(1) $-\,\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x-z)\cdot\mathcal{K}(y-z)\,f^{abc}\rho^d(x')\rho^h(y)\rho^e(x)U^{db}(w')U^{ec}(x)U^{ah}(y)$ (α)

(2) $+\,\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\rho^d(x')\rho^e(x)\rho^h(y)U^{dc}(x')U^{eb}(w)U^{ah}(y)$ (α')

(3) $-\,\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\rho^d(x')\rho^e(x)\rho^h(y)U^{dc}(x')U^{be}(x)U^{ah}(y)$ (α')

(4) $+\,\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x-z)\cdot\mathcal{K}(y-z)\,f^{abc}\rho^e(x')\rho^h(y)\rho^d(x)U^{dc}(x)U^{be}(x')U^{ah}(y)$ (α)

Group I, Row II, part 1

(1) $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-w)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^e(x')\,\rho^b(x)\,\rho^c(y)\,U^{de}(x')\,U^{ad}(w)$ (α) [removed, F4]

(2) $-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-w)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^e(x')\,\rho^b(y)\,\rho^c(x)\,U^{de}(x')\,U^{ad}(w)$ (α) [removed, F4]

(3) $-\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^d(x')\,\rho^e(x)\,\rho^h(y)\,U^{ad}(x')\,U^{be}(x)\,U^{ch}(y)$ (α) [T3]

(4) $+\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^d(x')\,\rho^e(y)\,\rho^h(x)\,U^{ad}(x')\,U^{be}(y)\,U^{ch}(x)$ (α) [T3]

(5) $-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-w)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^e(x')\,\rho^b(x)\,\rho^c(y)\,U^{de}(w')\,U^{ad}(w)$ (α) [removed, F4]

(6) $+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-w)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^e(x')\,\rho^b(y)\,\rho^c(x)\,U^{ed}(w')\,U^{ad}(w)$ (α) [removed, F4]

(7) $+\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^d(x')\,\rho^e(x)\,\rho^h(y)\,U^{ad}(w')\,U^{be}(x)\,U^{ch}(y)$ (α) [T3]

(8) $-\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^d(x')\,\rho^e(y)\,\rho^h(x)\,U^{ad}(w')\,U^{be}(y)\,U^{ch}(x)$ (α) [T3]

(9) $-\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^e(x')\,\rho^b(x)\,\rho^c(y)\,U^{de}(x')\,U^{ad}(w)$ (β)

(10) $+\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^e(x')\,\rho^b(y)\,\rho^c(x)\,U^{de}(x')\,U^{ad}(w)$ (β)

(11) $+\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^d(x')\,\rho^e(x)\,\rho^h(y)\,U^{ad}(x')\,U^{be}(x)\,U^{ch}(y)$ (β)

(12) $-\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^d(x')\,\rho^e(y)\,\rho^h(x)\,U^{ad}(x')\,U^{be}(y)\,U^{ch}(x)$ (β)

(13) $+\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^e(x')\,\rho^b(x)\,\rho^c(y)\,U^{de}(w')\,U^{ad}(w)$ (β)

(14) $-\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^e(x')\,\rho^b(y)\,\rho^c(x)\,U^{de}(w')\,U^{ad}(w)$ (β)

(15) $-\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^d(x')\,\rho^e(x)\,\rho^h(y)\,U^{ad}(w')\,U^{be}(x)\,U^{ch}(y)$ (β)

(16) $+\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^d(x')\,\rho^e(y)\,\rho^h(x)\,U^{ad}(w')\,U^{be}(y)\,U^{ch}(x)$ (β)

Group I, Row II, part 2 (c.c.)

(1) $-\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^{e}(x)\,\rho^{b}(x')\,\rho^{c}(y)\,U^{de}(x)\,U^{ad}(w')\;(\alpha')$ [removed, F4]

(2) $+\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^{e}(x)\,\rho^{b}(y)\,\rho^{c}(x')\,U^{de}(x)\,U^{ad}(w')\;(\alpha')$ [removed, F4]

(3) $+\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^{h}(y)\,\rho^{e}(x')\,\rho^{d}(x)\,U^{ad}(x)\,U^{be}(x')\,U^{ch}(y)\;(\alpha')$ [c.c.]

(4) $-\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^{h}(x')\,\rho^{e}(y)\,\rho^{d}(x)\,U^{ad}(x)\,U^{be}(y)\,U^{ch}(x')\;(\alpha')$ [c.c.]

(5) $+\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^{e}(x)\,\rho^{b}(x')\,\rho^{c}(y)\,U^{de}(w)\,U^{ad}(w')\;(\alpha')$ [removed, F4]

(6) $-\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^{e}(x)\,\rho^{b}(y)\,\rho^{c}(x')\,U^{de}(w)\,U^{ad}(w')\;(\alpha')$ [removed, F4]

(7) $-\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^{h}(y)\,\rho^{e}(x')\,\rho^{d}(x)\,U^{ad}(w)\,U^{be}(x')\,U^{ch}(y)\;(\alpha')$ [c.c.]

(8) $+\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^{h}(x')\,\rho^{e}(y)\,\rho^{d}(x)\,U^{ad}(w)\,U^{be}(y)\,U^{ch}(x')\;(\alpha')$ [c.c.]

(9) $+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^{c}(y)\,\rho^{b}(x')\,\rho^{e}(x)\,U^{de}(x)\,U^{ad}(w')\;(\beta')$ [c.c.]

(10) $-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^{c}(x')\,\rho^{b}(y)\,\rho^{e}(x)\,U^{de}(x)\,U^{ad}(w')\;(\beta')$ [c.c.]

(11) $-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^{h}(y)\,\rho^{e}(x')\,\rho^{d}(x)\,U^{ad}(x)\,U^{be}(x')\,U^{ch}(y)\;(\beta')$ [c.c.]

(12) $+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^{h}(x')\,\rho^{e}(y)\,\rho^{d}(x)\,U^{ad}(x)\,U^{be}(y)\,U^{ch}(x')\;(\beta')$ [c.c.]

(13) $-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^{c}(y)\,\rho^{b}(x')\,\rho^{e}(x)\,U^{de}(w)\,U^{ad}(w')\;(\beta')$ [c.c.]

(14) $+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^{c}(x')\,\rho^{b}(y)\,\rho^{e}(x)\,U^{de}(w)\,U^{ad}(w')\;(\beta')$ [c.c.]

(15) $+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^{h}(y)\,\rho^{e}(x')\,\rho^{d}(x)\,U^{ad}(w)\,U^{be}(x')\,U^{ch}(y)\;(\beta')$ [c.c.]

(16) $-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^{h}(x')\,\rho^{e}(y)\,\rho^{d}(x)\,U^{ad}(w)\,U^{be}(y)\,U^{ch}(x')\;(\beta')$ [c.c.]

Group I, Row III

(1) $\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x-z)\cdot\mathcal{K}(y-z)\,f^{abc}\rho^{d}(x')\rho^{e}(x)\rho^{h}(y)\,U^{ad}(x')\,U^{ce}(x)\,U^{bh}(z)\;(\alpha)$

(2) $-\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x-z)\cdot\mathcal{K}(y-z)\,f^{abc}\rho^{d}(x')\rho^{e}(x)\rho^{h}(y)\,U^{ad}(x')\,U^{ce}(x)\,U^{bh}(y)\;(\alpha)$

(3) $-\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x-z)\cdot\mathcal{K}(y-z)\,f^{abc}\rho^{d}(x')\rho^{e}(x)\rho^{h}(y)\,U^{ad}(w')\,U^{ce}(x)\,U^{bh}(z)\;(\alpha)$

(4) $+\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x-z)\cdot\mathcal{K}(y-z)\,f^{abc}\rho^{d}(x')\rho^{e}(x)\rho^{h}(y)\,U^{ad}(w')\,U^{ce}(x)\,U^{bh}(y)\;(\alpha)$

(5) $-\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^{h}(y)\,\rho^{e}(x')\,\rho^{d}(x)\,U^{ad}(x)\,U^{ce}(x')\,U^{bh}(z)\;(\alpha')$ [c.c.]

(6) $+\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^{h}(y)\,\rho^{e}(x')\,\rho^{d}(x)\,U^{ad}(x)\,U^{ce}(x')\,U^{bh}(y)\;(\alpha')$ [c.c.]

(7) $+\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^{h}(y)\,\rho^{e}(x')\,\rho^{d}(x)\,U^{ad}(w)\,U^{ce}(x')\,U^{bh}(z)\;(\alpha')$ [c.c.]

(8) $-\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^{h}(y)\,\rho^{e}(x')\,\rho^{d}(x)\,U^{ad}(w)\,U^{ce}(x')\,U^{bh}(y)\;(\alpha')$ [c.c.]

(9) $+2\,\mathcal{K}(x'-w')\cdot\mathcal{K}(y-z)\,\phi_{w}(x)\,f^{abc}\,\rho^{d}(x')\,\rho^{e}(x)\,\rho^{h}(y)\,U^{ad}(x')\,U^{ce}(x)\,U^{bh}(z)\;(\gamma)$ [removed, F3]

(10) $-2\,\mathcal{K}(x'-w')\cdot\mathcal{K}(y-z)\,\phi_{w}(x)\,f^{abc}\,\rho^{d}(x')\,\rho^{e}(x)\,\rho^{h}(y)\,U^{ad}(x')\,U^{ce}(x)\,U^{bh}(y)\;(\gamma)$ [removed, F3]

(11) $-2\,\mathcal{K}(x'-w')\cdot\mathcal{K}(y-z)\,\phi_{w}(x)\,f^{abc}\,\rho^{d}(x')\,\rho^{e}(x)\,\rho^{h}(y)\,U^{ad}(w')\,U^{ce}(x)\,U^{bh}(z)\;(\gamma)$ [removed, F3]

(12) $+2\,\mathcal{K}(x'-w')\cdot\mathcal{K}(y-z)\,\phi_{w}(x)\,f^{abc}\,\rho^{d}(x')\,\rho^{e}(x)\,\rho^{h}(y)\,U^{ad}(w')\,U^{ce}(x)\,U^{bh}(y)\;(\gamma)$ [removed, F3]

(13) $-2\,\mathcal{K}(x-w)\cdot\mathcal{K}(y-z)\,\phi_{w'}(x')\,f^{abc}\,\rho^{d}(x)\,\rho^{e}(x')\,\rho^{h}(y)\,U^{ad}(x)\,U^{ce}(x')\,U^{bh}(z)\;(\gamma')$ [removed, F3]

(14) $+2\,\mathcal{K}(x-w)\cdot\mathcal{K}(y-z)\,\phi_{w'}(x')\,f^{abc}\,\rho^{d}(x)\,\rho^{e}(x')\,\rho^{h}(y)\,U^{ad}(x)\,U^{ce}(x')\,U^{bh}(y)\;(\gamma')$ [removed, F3]

(15) $+2\,\mathcal{K}(x-w)\cdot\mathcal{K}(y-z)\,\phi_{w'}(x')\,f^{abc}\,\rho^{d}(x)\,\rho^{e}(x')\,\rho^{h}(y)\,U^{ad}(w)\,U^{ce}(x')\,U^{bh}(z)\;(\gamma')$ [removed, F3]

(16) $-2\,\mathcal{K}(x-w)\cdot\mathcal{K}(y-z)\,\phi_{w'}(x')\,f^{abc}\,\rho^{d}(x)\,\rho^{e}(x')\,\rho^{h}(y)\,U^{ad}(w)\,U^{ce}(x')\,U^{bh}(y)\;(\gamma')$ [removed, F3]

Group I, Row III: $\mathcal{D}$, replaces the γ,γ' terms

(D1) $-2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^{d}(x')\,\rho^{e}(y)\,\rho^{h}(x)\,U^{ad}(x')\,U^{ce}(y)\,U^{bh}(w)\;(\beta)$ [added, F3]

(D2) $+2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^{d}(x')\,\rho^{e}(y)\,\rho^{h}(x)\,U^{ad}(x')\,U^{ce}(y)\,U^{bh}(x)\;(\beta)$ [added, F3]

(D3) $+2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^{d}(x')\,\rho^{e}(y)\,\rho^{h}(x)\,U^{ad}(w')\,U^{ce}(y)\,U^{bh}(w)\;(\beta)$ [added, F3]

(D4) $-2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^{d}(x')\,\rho^{e}(y)\,\rho^{h}(x)\,U^{ad}(w')\,U^{ce}(y)\,U^{bh}(x)\;(\beta)$ [added, F3]

(D5) $+2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^{h}(x')\,\rho^{e}(y)\,\rho^{d}(x)\,U^{ad}(x)\,U^{ce}(y)\,U^{bh}(w')\;(\beta')$ [added, F3]

(D6) $-2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^{h}(x')\,\rho^{e}(y)\,\rho^{d}(x)\,U^{ad}(x)\,U^{ce}(y)\,U^{bh}(x')\;(\beta')$ [added, F3]

(D7) $-2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^{h}(x')\,\rho^{e}(y)\,\rho^{d}(x)\,U^{ad}(w)\,U^{ce}(y)\,U^{bh}(w')\;(\beta')$ [added, F3]

(D8) $+2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^{h}(x')\,\rho^{e}(y)\,\rho^{d}(x)\,U^{ad}(w)\,U^{ce}(y)\,U^{bh}(x')\;(\beta')$ [added, F3]

Group I, Row IV

(1) $\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^{e}(x')\,\rho^{l}(y)\,\rho^{c}(x)\,U^{de}(x')\,U^{ad}(w)\,U^{hb}(z)\,U^{hl}(y)\;(\alpha)$

(2) $-\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^{d}(x')\,\rho^{e}(y)\,\rho^{h}(x)\,U^{da}(x')\,U^{be}(y)\,U^{ch}(x)\;(\alpha)$

(3) $-\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^{e}(x')\,\rho^{l}(y)\,\rho^{c}(x)\,U^{de}(w')\,U^{ad}(w)\,U^{hb}(z)\,U^{hl}(y)\;(\alpha)$

(4) $+\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^{d}(x')\,\rho^{e}(y)\,\rho^{h}(x)\,U^{da}(w')\,U^{be}(y)\,U^{ch}(x)\;(\alpha)$

(5) $-\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^{c}(x')\,\rho^{l}(y)\,\rho^{e}(x)\,U^{de}(x)\,U^{ad}(w')\,U^{hb}(z)\,U^{hl}(y)\;(\alpha')$ [c.c.]

(6) $+\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^{h}(x')\,\rho^{e}(y)\,\rho^{d}(x)\,U^{da}(x)\,U^{ch}(x')\,U^{be}(y)\;(\alpha')$ [c.c.]

(7) $+\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^{c}(x')\,\rho^{l}(y)\,\rho^{e}(x)\,U^{de}(w)\,U^{ad}(w')\,U^{hb}(z)\,U^{hl}(y)\;(\alpha')$ [c.c.]

(8) $-\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^{h}(x')\,\rho^{e}(y)\,\rho^{d}(x)\,U^{da}(w)\,U^{ch}(x')\,U^{be}(y)\;(\alpha')$ [c.c.]

(9) $-2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^{e}(x')\,\rho^{l}(y)\,\rho^{c}(x)\,U^{de}(x')\,U^{ad}(w)\,U^{hb}(z)\,U^{hl}(y)\;(\beta)$

(10) $+2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^{e}(x')\,\rho^{l}(y)\,\rho^{c}(x)\,U^{de}(w')\,U^{ad}(w)\,U^{hb}(z)\,U^{hl}(y)\;(\beta)$

(11) $+2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^{d}(x')\,\rho^{e}(y)\,\rho^{h}(x)\,U^{da}(x')\,U^{be}(y)\,U^{hc}(x)\;(\beta)$

(12) $-2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^{d}(x')\,\rho^{e}(y)\,\rho^{h}(x)\,U^{da}(w')\,U^{be}(y)\,U^{hc}(x)\;(\beta)$

(13) $+2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^{c}(x')\,\rho^{l}(y)\,\rho^{e}(x)\,U^{de}(x)\,U^{ad}(w')\,U^{hb}(z)\,U^{hl}(y)\;(\beta')$ [c.c.]

(14) $-2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^{c}(x')\,\rho^{l}(y)\,\rho^{e}(x)\,U^{de}(w)\,U^{ad}(w')\,U^{hb}(z)\,U^{hl}(y)\;(\beta')$ [c.c.]

(15) $+2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^{h}(x')\,\rho^{e}(y)\,\rho^{d}(x)\,U^{be}(y)\,U^{ad}(w)\,U^{ch}(x')\;(\beta')$ [c.c.]

(16) $-2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^{h}(x')\,\rho^{e}(y)\,\rho^{d}(x)\,U^{be}(y)\,U^{ad}(x)\,U^{ch}(x')\;(\beta')$ [c.c.]

Group I, Row V, part 1

(1) $-\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^{d}(x')\{\rho^{h}(x),\rho^{e}(y)\}\,U^{ad}(x')\,U^{be}(z)\,U^{ch}(w)\;(\beta)$ [order]

(2) $+\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^{d}(x')\{\rho^{h}(x),\rho^{e}(y)\}\,U^{ad}(w')\,U^{be}(z)\,U^{ch}(w)\;(\beta)$ [order]

(3) $+\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^{d}(x')\{\rho^{e}(x),\rho^{h}(y)\}\,U^{ad}(x')\,U^{ce}(x)\,U^{bh}(y)\;(\beta)$ [order]

(4) $-\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^{d}(x')\{\rho^{e}(x),\rho^{h}(y)\}\,U^{ad}(w')\,U^{ce}(x)\,U^{bh}(y)\;(\beta)$ [order]

(5) $+\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\{\rho^{h}(x'),\rho^{e}(y)\}\rho^{d}(x)\,U^{ad}(x)\,U^{be}(z)\,U^{ch}(w')\;(\beta')$ [order]

(6) $-\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\{\rho^{h}(x'),\rho^{e}(y)\}\rho^{d}(x)\,U^{ad}(w)\,U^{be}(z)\,U^{ch}(w')\;(\beta')$ [order]

(7) $+\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\{\rho^{e}(x'),\rho^{h}(y)\}\rho^{d}(x)\,U^{ad}(w)\,U^{bh}(y)\,U^{ce}(x')\;(\beta')$ [order]

(8) $-\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\{\rho^{e}(x'),\rho^{h}(y)\}\rho^{d}(x)\,U^{ad}(x)\,U^{bh}(y)\,U^{ce}(x')\;(\beta')$ [order]

Group I, Row V, part 2

(1) $+\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^{d}(x')\{\rho^{e}(x),\rho^{h}(y)\}\,U^{ad}(x')\,U^{be}(w)\,U^{ch}(z)\;(\beta)$ [order]

(2) $-\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^{d}(x')\{\rho^{h}(x),\rho^{e}(y)\}\,U^{ad}(x')\,U^{ce}(y)\,U^{bh}(x)\;(\beta)$ [order]

(3) $-\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^{d}(x')\{\rho^{e}(x),\rho^{h}(y)\}\,U^{ad}(w')\,U^{be}(w)\,U^{ch}(z)\;(\beta)$ [order]

(4) $+\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^{d}(x')\{\rho^{h}(x),\rho^{e}(y)\}\,U^{ad}(w')\,U^{ce}(y)\,U^{bh}(x)\;(\beta)$ [order]

(5) $-\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\{\rho^{e}(x'),\rho^{h}(y)\}\rho^{d}(x)\,U^{ad}(x)\,U^{be}(w')\,U^{ch}(z)\;(\beta')$ [order]

(6) $+\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\{\rho^{h}(x'),\rho^{e}(y)\}\rho^{d}(x)\,U^{ad}(x)\,U^{ce}(y)\,U^{bh}(x')\;(\beta')$ [order]

(7) $+\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\{\rho^{e}(x'),\rho^{h}(y)\}\rho^{d}(x)\,U^{ad}(w)\,U^{ch}(z)\,U^{be}(w')\;(\beta')$ [order]

(8) $-\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\{\rho^{h}(x'),\rho^{e}(y)\}\rho^{d}(x)\,U^{ad}(w)\,U^{bh}(x')\,U^{ce}(y)\;(\beta')$ [order]

Group II, Row I

(1) $-\frac{1}{2}\,\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\{\rho^{e}(x'),\rho^{b}(y)\}\rho^{c}(x)\,U^{de}(w')\,U^{ad}(w)\;(\alpha)$ [order]

(2) $+\frac{1}{2}\,\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\{\rho^{d}(x'),\rho^{e}(y)\}\rho^{h}(x)\,U^{be}(z)\,U^{ad}(w')\,U^{ch}(x)\;(\alpha)$ [order]

(3) $+\frac{1}{2}\,\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\{\rho^{h}(x'),\rho^{l}(y)\}\rho^{c}(x)\,U^{eh}(x')\,U^{bd}(z)\,U^{dl}(y)\,U^{ae}(w)\;(\alpha)$ [order]

(4) $-\frac{1}{2}\,\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\{\rho^{d}(x'),\rho^{e}(y)\}\rho^{h}(x)\,U^{ad}(x')\,U^{be}(y)\,U^{ch}(x)\;(\alpha)$ [order]

(5) $+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\{\rho^{e}(x'),\rho^{b}(y)\}\rho^{c}(x)\,U^{de}(w')\,U^{ad}(w)\;(\beta)$ [order]

(6) $-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\{\rho^{d}(x'),\rho^{e}(y)\}\rho^{h}(x)\,U^{ad}(w')\,U^{eb}(z)\,U^{ch}(x)\;(\beta)$ [order]

(7) $-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\{\rho^{h}(x'),\rho^{l}(y)\}\rho^{c}(x)\,U^{eh}(x')\,U^{bd}(z)\,U^{dl}(y)\,U^{ea}(w)\;(\beta)$ [order]

(8) $+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\{\rho^{d}(x'),\rho^{e}(y)\}\rho^{h}(x)\,U^{da}(x')\,U^{be}(y)\,U^{ch}(x)\;(\beta)$ [order]

(9) $+\frac{1}{2}\,\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^{c}(x')\{\rho^{e}(x),\rho^{b}(y)\}\,U^{ad}(w')\,U^{de}(w)\;(\alpha')$ [order]

(10) $-\frac{1}{2}\,\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^{c}(x')\{\rho^{h}(x),\rho^{l}(y)\}\,U^{ad}(w')\,U^{be}(z)\,U^{dh}(x)\,U^{el}(y)\;(\alpha')$ [order]

(11) $-\frac{1}{2}\,\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^{e}(x')\{\rho^{h}(x),\rho^{d}(y)\}\,U^{bd}(z)\,U^{ce}(x')\,U^{ah}(w)\;(\alpha')$ [order]

(12) $+\frac{1}{2}\,\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^{d}(x')\{\rho^{e}(x),\rho^{h}(y)\}\,U^{cd}(x')\,U^{ae}(x)\,U^{bh}(y)\;(\alpha')$ [order]

(13) $-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^{c}(x')\{\rho^{e}(x),\rho^{b}(y)\}\,U^{ad}(w')\,U^{de}(w)\;(\beta')$ [order]

(14) $+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^{c}(x')\{\rho^{h}(x),\rho^{l}(y)\}\,U^{ad}(w')\,U^{be}(z)\,U^{dh}(x)\,U^{el}(y)\;(\beta')$ [order]

(15) $+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^{e}(x')\{\rho^{h}(x),\rho^{d}(y)\}\,U^{ce}(x')\,U^{db}(z)\,U^{ah}(w)\;(\beta')$ [order]

(16) $-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^{d}(x')\{\rho^{e}(x),\rho^{h}(y)\}\,U^{cd}(x')\,U^{ae}(x)\,U^{bh}(y)\;(\beta')$ [order]

Group III, Row I

(1) $-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^{e}(x')\,\rho^{l}(y)\,\rho^{c}(x)\,U^{de}(x')\,U^{hl}(y)\,U^{hb}(z)\,U^{ad}(w)\;(\alpha)$

(2) $+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^{d}(x')\,\rho^{e}(y)\,\rho^{h}(x)\,U^{ad}(x')\,U^{be}(y)\,U^{ch}(x)\;(\alpha)$

(3) $+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^{l}(x')\,\rho^{e}(y)\,\rho^{c}(x)\,U^{bd}(z)\,U^{de}(y)\,U^{hl}(w')\,U^{ah}(w)$ (α)

(4) $-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^{e}(x')\,\rho^{d}(y)\,\rho^{h}(x)\,U^{ae}(w')\,U^{bd}(y)\,U^{ch}(x)$ (α)

(5) $+2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^{e}(x')\,\rho^{l}(y)\,\rho^{c}(x)\,U^{de}(x')\,U^{hl}(y)\,U^{hb}(z)\,U^{ad}(w)$ (β)

(6) $-2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^{d}(x')\,\rho^{e}(y)\,\rho^{h}(x)\,U^{ad}(x')\,U^{be}(y)\,U^{ch}(x)$ (β)

(7) $-2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^{l}(x')\,\rho^{e}(y)\,\rho^{c}(x)\,U^{bd}(z)\,U^{de}(y)\,U^{hl}(w')\,U^{ah}(w)$ (β)

(8) $+2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^{e}(x')\,\rho^{d}(y)\,\rho^{h}(x)\,U^{ae}(w')\,U^{bd}(y)\,U^{ch}(x)$ (β)

(9) $+\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^{c}(x')\,\rho^{l}(y)\,\rho^{e}(x)\,U^{de}(x)\,U^{hl}(y)\,U^{hb}(z)\,U^{ad}(w')$ (α') [c.c.]

(10) $-\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^{h}(x')\,\rho^{e}(y)\,\rho^{d}(x)\,U^{ad}(x)\,U^{be}(y)\,U^{ch}(x')$ (α') [c.c.]

(11) $-\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^{c}(x')\,\rho^{e}(y)\,\rho^{l}(x)\,U^{bd}(z)\,U^{de}(y)\,U^{hl}(w)\,U^{ha}(w')$ (α') [c.c.]

(12) $+\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^{h}(x')\,\rho^{d}(y)\,\rho^{e}(x)\,U^{bd}(y)\,U^{ae}(w)\,U^{ch}(x')$ (α') [c.c.]

(13) $-2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^{c}(x')\,\rho^{l}(y)\,\rho^{e}(x)\,U^{de}(x)\,U^{hl}(y)\,U^{hb}(z)\,U^{ad}(w')$ (β') [c.c.]

(14) $+2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^{h}(x')\,\rho^{e}(y)\,\rho^{d}(x)\,U^{ad}(x)\,U^{be}(y)\,U^{ch}(x')$ (β') [c.c.]

(15) $+2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^{c}(x')\,\rho^{e}(y)\,\rho^{l}(x)\,U^{de}(y)\,U^{hl}(w)\,U^{db}(z)\,U^{ha}(w')$ (β') [T1, c.c.]

(16) $-2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^{h}(x')\,\rho^{d}(y)\,\rho^{e}(x)\,U^{bd}(y)\,U^{ae}(w)\,U^{ch}(x')$ (β') [c.c.]

Group III, Row II

(1) $-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^{e}(y)\,\rho^{l}(x')\,\rho^{c}(x)\,U^{de}(y)\,U^{hl}(x')\,U^{bd}(z)\,U^{ah}(w)$ (α)

(2) $+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^{d}(y)\,\rho^{e}(x')\,\rho^{h}(x)\,U^{bd}(y)\,U^{ae}(x')\,U^{ch}(x)$ (α)

(3) $+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^{b}(y)\,\rho^{e}(x')\,\rho^{c}(x)\,U^{de}(x')\,U^{ad}(w)$ (α)

(4) $-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^{e}(y)\,\rho^{d}(x')\,\rho^{h}(x)\,U^{ad}(x')\,U^{eb}(z)\,U^{ch}(x)$ (α)

(5) $+2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^{e}(y)\,\rho^{l}(x')\,\rho^{c}(x)\,U^{de}(y)\,U^{hl}(x')\,U^{bd}(z)\,U^{ha}(w)$ (β)

(6) $-2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^{d}(y)\,\rho^{e}(x')\,\rho^{h}(x)\,U^{db}(y)\,U^{ae}(x')\,U^{ch}(x)$ (β)

(7) $-2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^{b}(y)\,\rho^{e}(x')\,\rho^{c}(x)\,U^{de}(x')\,U^{da}(w)$ (β)

(8) $+2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^{e}(y)\,\rho^{d}(x')\,\rho^{h}(x)\,U^{ad}(x')\,U^{eb}(z)\,U^{ch}(x)$ (β)

(9) $+\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^{c}(x')\,\rho^{l}(x)\,\rho^{e}(y)\,U^{de}(y)\,U^{hl}(x)\,U^{bd}(z)\,U^{ah}(w')$ (α') [c.c.]

(10) $-\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^{h}(x')\,\rho^{e}(x)\,\rho^{d}(y)\,U^{bd}(y)\,U^{ae}(x)\,U^{ch}(x')$ (α') [c.c.]

(11) $-\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^{c}(x')\,\rho^{e}(x)\,\rho^{b}(y)\,U^{de}(x)\,U^{ad}(w')$ (α') [c.c.]

(12) $+\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^{h}(x')\,\rho^{d}(x)\,\rho^{e}(y)\,U^{ad}(x)\,U^{eb}(z)\,U^{ch}(x')$ (α') [c.c.]

(13) $-2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^{c}(x')\,\rho^{l}(x)\,\rho^{e}(y)\,U^{de}(y)\,U^{hl}(x)\,U^{bd}(z)\,U^{ha}(w')$ (β') [T2, c.c.]

(14) $+2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^{h}(x')\,\rho^{e}(x)\,\rho^{d}(y)\,U^{bd}(y)\,U^{ae}(x)\,U^{ch}(x')$ (β') [c.c.]

(15) $+2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^{c}(x')\,\rho^{e}(x)\,\rho^{b}(y)\,U^{de}(x)\,U^{ad}(w')$ (β') [c.c.]

(16) $-2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^{h}(x')\,\rho^{d}(x)\,\rho^{e}(y)\,U^{ad}(x)\,U^{eb}(z)\,U^{ch}(x')$ (β') [c.c.]

Group III, Row III

(1) $+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\{\rho^{h}(x'),\rho^{e}(y)\}\rho^{d}(x)\,U^{ad}(w)\,U^{be}(z)\,U^{ch}(w')$ (β) [order]

(2) $-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\{\rho^{e}(x'),\rho^{h}(y)\}\rho^{d}(x)\,U^{ad}(w)\,U^{ce}(x')\,U^{bh}(y)$ (β) [order]

(3) $-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\{\rho^{h}(x'),\rho^{d}(y)\}\rho^{e}(x)\,U^{db}(z)\,U^{ae}(x')\,U^{ch}(w')$ (β) [order]

(4) $+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\{\rho^{e}(x'),\rho^{h}(y)\}\rho^{d}(x)\,U^{ad}(x)\,U^{ce}(x')\,U^{bh}(y)$ (β) [order]

(5) $-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^{d}(x')\{\rho^{h}(x),\rho^{e}(y)\}\,U^{ad}(w')\,U^{be}(z)\,U^{ch}(w)$ (β') [order]

(6) $+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^{d}(x')\{\rho^{e}(x),\rho^{h}(y)\}\,U^{ad}(w')\,U^{ce}(x)\,U^{bh}(y)$ (β') [order]

(7) $+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^{e}(x')\{\rho^{h}(x),\rho^{d}(y)\}\,U^{bd}(z)\,U^{ae}(x')\,U^{ch}(w)$ (β') [order]

(8) $-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^{d}(x')\{\rho^{e}(x),\rho^{h}(y)\}\,U^{ad}(x')\,U^{ce}(x)\,U^{bh}(y)$ (β') [order]

Group III, Row V

(1) $+2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^{h}(x')\,\rho^{e}(y)\,\rho^{d}(x)\,U^{ad}(w)\,U^{be}(y)\,U^{ch}(x')$ (β) [c.c.]

(2) $-2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^{e}(x')\,\rho^{h}(y)\,\rho^{d}(x)\,U^{ad}(w)\,U^{ce}(w')\,U^{bh}(y)$ (β) [c.c.]

(3) $-2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^{h}(x')\,\rho^{e}(y)\,\rho^{d}(x)\,U^{be}(y)\,U^{ad}(x)\,U^{ch}(x')$ (β) [c.c.]

(4) $+2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^{e}(x')\,\rho^{h}(y)\,\rho^{d}(x)\,U^{ad}(x)\,U^{ce}(w')\,U^{bh}(y)$ (β) [c.c.]

(5) $-2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^{d}(x')\,\rho^{e}(y)\,\rho^{h}(x)\,U^{ad}(w')\,U^{be}(y)\,U^{ch}(x)$ (β')

(6) $+2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^{d}(x')\,\rho^{h}(y)\,\rho^{e}(x)\,U^{ad}(w')\,U^{ce}(w)\,U^{bh}(y)$ (β')

(7) $+2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^{d}(x')\,\rho^{e}(y)\,\rho^{h}(x)\,U^{be}(y)\,U^{ad}(x')\,U^{ch}(x)$ (β')

(8) $-2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^{d}(x')\,\rho^{h}(y)\,\rho^{e}(x)\,U^{ad}(x')\,U^{ce}(w)\,U^{bh}(y)$ (β')

Group III, Row VI

$$
\begin{aligned}
(1)\quad & +2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^{h}(y)\,\rho^{e}(x')\,\rho^{d}(x)\,U^{ad}(w)\,U^{ce}(x')\,U^{bh}(y)\;(\beta) \quad \text{[c.c.]}\\
(2)\quad & -2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^{e}(y)\,\rho^{h}(x')\,\rho^{d}(x)\,U^{ad}(w)\,U^{be}(z)\,U^{ch}(x')\;(\beta) \quad \text{[c.c.]}\\
(3)\quad & -2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^{h}(y)\,\rho^{e}(x')\,\rho^{d}(x)\,U^{bh}(y)\,U^{ad}(x)\,U^{ce}(x')\;(\beta) \quad \text{[c.c.]}\\
(4)\quad & +2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^{d}(y)\,\rho^{h}(x')\,\rho^{e}(x)\,U^{ae}(x)\,U^{ch}(x')\,U^{db}(z)\;(\beta) \quad \text{[c.c.]}\\
(5)\quad & -2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^{d}(x')\,\rho^{e}(x)\,\rho^{h}(y)\,U^{ad}(w')\,U^{bh}(y)\,U^{ce}(x)\;(\beta')\\
(6)\quad & +2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^{d}(x')\,\rho^{h}(x)\,\rho^{e}(y)\,U^{ad}(w')\,U^{be}(z)\,U^{ch}(x)\;(\beta')\\
(7)\quad & +2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^{d}(x')\,\rho^{e}(x)\,\rho^{h}(y)\,U^{bh}(y)\,U^{ad}(x')\,U^{ce}(x)\;(\beta')\\
(8)\quad & -2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^{e}(x')\,\rho^{h}(x)\,\rho^{d}(y)\,U^{ae}(x')\,U^{ch}(x)\,U^{bd}(z)\;(\beta')\Big]
\end{aligned}
\tag{1}
$$